Olympiad Geometry Set

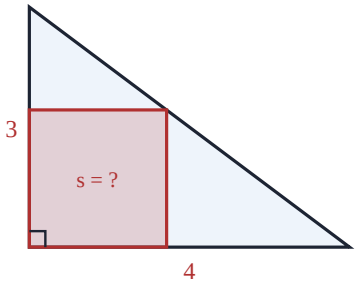
Unit 3 topics Similar triangles · volume · surface area · scale 10 problems

Olympiad-style problems of the kind used in contests such as AMC 8/10, Math Kangaroo and the CEMC Pascal/Cayley contests, including several well-known classics. They are written in that style, not copied from a specific past paper. Answers with full solutions are at the end.

Similar Triangles

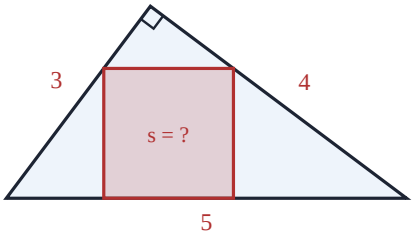
Problem 1

A right triangle has legs 3 and 4. A square is drawn inside it so that two of its sides lie along the legs and one corner touches the hypotenuse. Find the side of the square.



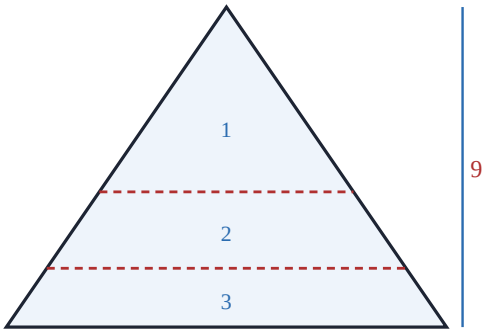
Problem 2

The same 3-4-5 triangle now has a square with one whole side lying on the hypotenuse and its other two corners on the legs. Find the side of this square. Which square is bigger, this one or the one in Problem 1?



Problem 3

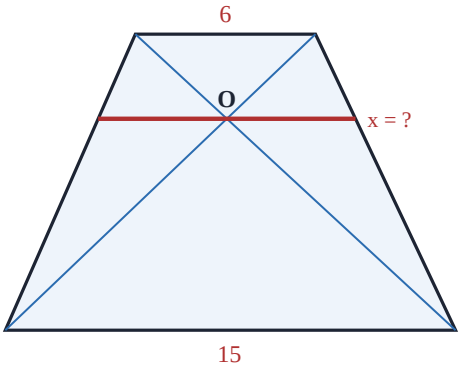
A triangle has height 9. Two lines parallel to its base cut it into three regions of equal area. How far from the top vertex is each line?



regions 1, 2, 3 have equal area

Problem 4

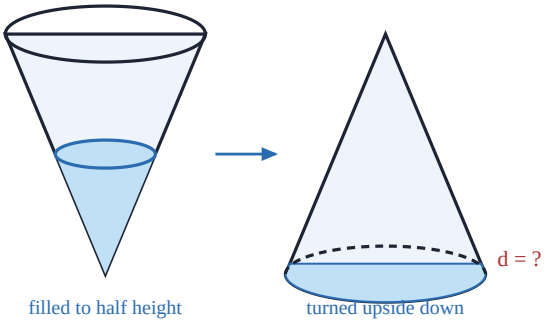
Trapezoid ABCD has parallel sides AB = 6 and DC = 15. A segment through O, where the diagonals cross, is drawn parallel to the bases from one leg to the other. Find its length x.



Volume & Similar Solids

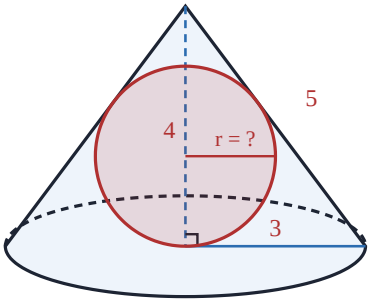
Problem 5

A cone-shaped glass (point down) is filled with water to half its height. (a) What fraction of the glass is full? (b) The glass is sealed and turned upside down, so the point is at the top. How deep is the water now, as a fraction of the glass's height?



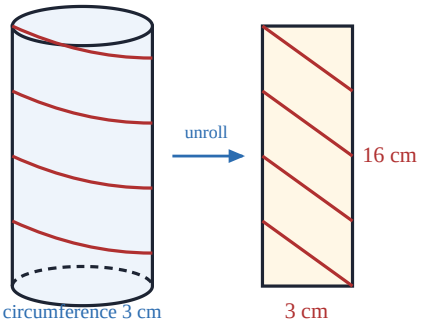
Problem 6

A sphere fits snugly inside a cone of base radius 3 and height 4, touching the base and the curved surface. Find the radius of the sphere.



Problem 7

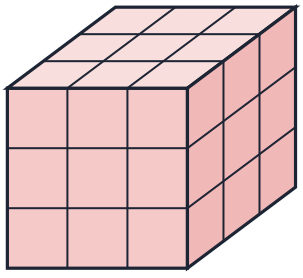
A string is wound evenly around a cylinder exactly 4 times, from the bottom to the top. The cylinder has circumference 3 cm and height 16 cm. How long is the string?



Surface Area & Counting

Problem 8

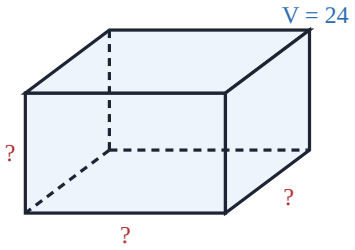
A $3 \times 3 \times 3$ cube is painted on all six faces and then cut into 27 small cubes. How many small cubes have exactly 3, 2, 1 and 0 painted faces? What are the answers for a $4 \times 4 \times 4$ cube?



painted, then cut into 1×1×1 cubes

Problem 9

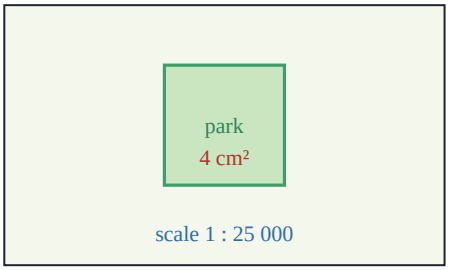
A box has whole-number edge lengths and a volume of 24 cm^3. Which dimensions give the smallest surface area, and what is it?



Scale

Problem 10

On a map with scale 1 : 25 000, a square park has an area of 4 cm^2. How long is each side of the real park, and what is its area in hectares?



SOLUTIONS

Similar Triangles

P1 Put the right angle at the origin. The small triangle above the square is similar to the whole triangle: $(3 - s)/s = 3/4 \Rightarrow 12 - 4s = 3s \Rightarrow \mathbf{s = 12/7 \approx 1.71}$. In general $s = ab/(a + b)$.

P2 The altitude to the hypotenuse is $h = 3 \cdot 4/5 = 2.4$. The triangle above the square is similar to the whole triangle with ratio $(h - s)/h$, so its base s satisfies $s/5 = (2.4 - s)/2.4 \Rightarrow 2.4s = 12 - 5s \Rightarrow \mathbf{s = 12/7.4 = 60/37 \approx 1.62}$. The square in Problem 1 is bigger ($12/7 \approx 1.71$).

P3 The triangle above a line at distance d is similar to the whole triangle with $k = d/9$, so its area is $(d/9)^2$ of the total. Equal thirds: $(d/9)^2 = 1/3$ and $2/3 \Rightarrow \mathbf{d = 9/\sqrt{3} \approx 5.20}$ and $\mathbf{d = 9\sqrt{(2/3)} \approx 7.35}$.

P4 $\Delta AOB \sim \Delta COD$ with ratio 6 : 15, so O divides each diagonal in the ratio 6 : 15 = 2 : 5. Using ΔADC (Thales): the part of the segment from the left leg to O is $15 \times 2/7 = 30/7$. Using ΔBDC: from O to the right leg is also $30/7$. So $\mathbf{x = 60/7 \approx 8.57}$. In general $x = 2ab/(a + b)$.

Volume & Similar Solids

P5 (a) The water is a cone similar to the glass with $k = ½ \Rightarrow$ volume $(½)^3 = \mathbf{1/8}$. (b) Upside down, the air at the top is a cone of volume 7/8 of the glass, so its height is $H \cdot \sqrt[3]{(7/8)} \approx 0.9565H$. The water depth is $\mathbf{H(1 - \sqrt[3]{(7/8)}) \approx 0.044H}$, only about 4% of the height.

P6 Cut through the axis: the cone becomes a triangle with base 6 and sides 5, 5 (slant $= \sqrt{(3^2 + 4^2)} = 5$). The sphere becomes the circle inside this triangle. Area $= ½ \cdot 6 \cdot 4 = 12$ and half the perimeter $= (5 + 5 + 6)/2 = 8$. Inscribed radius = area ÷ half-perimeter = $\mathbf{12/8 = 1.5}$. (Check with similar triangles: $(4 - r)/r = 5/3 \Rightarrow r = 1.5$.)

P7 Unroll the side of the cylinder into a rectangle 3 cm wide and 16 cm tall. Each turn becomes a straight diagonal across a 3 cm × 4 cm strip: $\sqrt{(3^2 + 4^2)} = 5$ cm. Four turns: **20 cm**. (Or one long diagonal: $\sqrt{(12^2 + 16^2)} = 20$.)

Surface Area & Counting

P8 3 × 3 × 3: corners have 3 faces painted: **8**; edge middles have 2: 12 edges × 1 = **12**; face centres have 1: 6 × 1 = **6**; the centre has 0: **1** (8 + 12 + 6 + 1 = 27 ✓). 4 × 4 × 4: **8, 24, 24, 8** (12 × 2, 6 × 4, 2^3).

P9 List the options: 1×1×24 → 98; 1×2×12 → 76; 1×3×8 → 70; 1×4×6 → 68; 2×2×6 → 56; **2×3×4 → 52 cm²**. The closer the box is to a cube, the smaller its surface area.

Scale

P10 Map side = 2 cm ⇒ real side = 2 × 25 000 = 50 000 cm = **500 m**. Area = 500^2 = 250 000 m^2 = **25 ha**. (Or 4 × $25\,000^2$ cm^2 = 2.5 × 10^9 cm^2 = 250 000 m^2.)